# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 8 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 7 & 8: **Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The change of liquid to vapour slowly below boiling point is: 1

**(a)** compression **(b)** melting **(c)** evaporation **(d)** freezing

**2.** Gases mix easily because their particles: 1

**(a)** are free and widely spaced **(b)** are fixed

**(c)** do not move **(d)** are closely packed

**3.** When ice melts, it changes from: 1

**(a)** liquid to gas **(b)** solid to gas directly

**(c)** solid to liquid **(d)** gas to solid

**4.** Which of the following can flow? 1

**(a)** iron only **(b)** ice only **(c)** liquid and gas **(d)** solid only

**5.** Increasing temperature of a gas generally makes its particles: 1

**(a)** stop **(b)** move faster **(c)** move slower **(d)** stick

**6.** On heating iron filings and sulfur, the product is: 1

**(a)** sulfur dioxide only **(b)** iron oxide

**(c)** salt **(d)** iron sulfide

**7.** Silicon and boron are called: 1

**(a)** metalloids **(b)** alkali metals **(c)** noble gases **(d)** alloys

**8.** Gold can be called a mineral and a metal because it is: 1

**(a)** a naturally occurring element that is a metal

**(b)** only a mixture

**(c)** only a compound

**(d)** an alloy

**9.** Which set has only compounds? 1

**(a)** Water, oxygen, salt **(b)** Air, sea water, brass

**(c)** Iron, sulfur, gold **(d)** Water, carbon dioxide, sodium chloride

**10.** Which statement is correct about compounds? 1

**(a)** They can be separated by filtration
**(b)** They have variable composition
**(c)** They are always mixtures
**(d)** They have properties different from their elements

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 10–15 words each.*

**11.** Why is the melting point of iron higher than that of ice? 2

**12.** What is evaporation? 2

**13.** What are minerals? 2

**14.** What are native minerals? 2

## SECTION — C (Short Answer Type) 4 × 3 = 12

*Answer in about 30–40 words each.*

**15.** State why the boiling of water at 100 °C is not at room temperature. 3

**16.** Why is milk classified as a mixture? 3

**17.** What happens to the interparticle space when a gas is cooled? Explain. 3

OR

Explain how particle theory supports that smell of camphor spreads in a room. 3

**18.** Explain with an example why a mixture does not have a fixed composition. 3

OR

Which has more uses, alloys or pure metals? Give reasons. 3

## SECTION — D (Long Answer Type) 2 × 5 = 10

*Answer in about 80–100 words each.*

**19.** Explain gas pressure and temperature relation using particle ideas. 5

OR

Discuss how particle ideas explain evaporation and boiling, and their differences. 5

**20.** Explain why compounds have properties different from their elements using sodium chloride and water. 5

OR

Discuss the importance of elements, compounds and mixtures in daily life with five examples. 5

## END OF QUESTION PAPER Total: 40 Marks

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